

KB-008-Stuck-Cable

KB-008 — Stuck / Locked Cable

Category: Physical Cable Issues

Call Volume: ~4% of contacts

Urgency: HIGH — driver is physically unable to leave

Primary Remote Actions: Remote connector unlock, technician dispatch

Why This Matters

A stuck cable is one of the most stressful situations an EV driver can face — their vehicle is physically tethered to a charger they cannot leave. Resolution needs to be fast and clear. Importantly, agents must never advise forcing the cable out: the CCS and CHAdeMO connectors have mechanical locking latches, and forcing them can cause expensive damage to the vehicle's charging inlet (costs can exceed \$1,000–\$3,000 to repair).

Background: Why Cables Lock and Stick

The Locking Mechanism

DC fast charging connectors (CCS and CHAdeMO) engage a mechanical latch when a session is active. This latch serves two purposes: 1. **Safety** — prevents accidental disconnection during high-voltage DC charging 2. **Integrity** — ensures the connection remains stable throughout the session

The latch is controlled by: - The charger's internal connector lock mechanism (a small motor or solenoid) - The vehicle's inlet latch (some vehicles have a secondary lock, controlled by the vehicle's charging controller)

Both must release for the connector to come out. If either stays locked, the cable is stuck.

Why a Cable Gets Stuck

Cause	Frequency	Notes
Session still technically active (or ghost session)	~40%	Easiest fix — end session, then unlock
Charger software failed to send unlock signal at session end	~25%	Remote unlock resolves
Vehicle inlet latch did not release (vehicle-side)	~15%	Vehicle manual release or dealer
Mechanical failure in charger lock mechanism	~10%	Remote unlock may not work — dispatch
Power loss to charger during active session	~5%	Manual release on charger or dispatch
Connector deformed or wedged due to prior misuse	~5%	Do not force — dispatch

Diagnostic Script

Step 1 — Confirm the scenario

“Is the cable stuck in your car’s port, or stuck at the charger end?”

Almost always: stuck in the vehicle’s charging port (the connector won’t pull out of the car).

Step 2 — Confirm session status

“What does the charger screen show right now — does it say the session is still active, or has it ended?”

“What does your vehicle’s display show — is it still showing a charging indicator?”

If session is still active:

“The connector is locked because the session is still running — that’s actually the latch doing its job. Let me help you end the session properly first.”

End session via: 1. Ask driver to tap “Stop Session” on the charger screen 2. Or stop via the network app 3. If neither works, agent triggers a remote stop via the backend

After the session ends, the connector should release. Ask driver to try again.

If session shows as ended but cable is still stuck:

“I see the session has ended on our side too. The connector should have released — let me send a remote unlock command to the station. Give it about 10 seconds after I send it, then try the cable.”

Trigger remote connector unlock. Wait 10–15 seconds.

“Okay, try the cable now — use a firm but smooth pull, don’t yank it.”

Step 3 — If remote unlock succeeds

“Did it release? Perfect. Make sure the cable is fully out before driving — and let me know if there’s anything else I can help with.”

Step 4 — If remote unlock fails

“The remote unlock didn’t work — the locking mechanism may have a hardware fault. Let’s try a couple of alternatives before we dispatch a technician.”

Vehicle-side manual release (check vehicle-specific options below):

Consult the vehicle-specific notes in Section B below.

Power cycle the charger: > “I’m going to try a full power cycle on the station — sometimes that releases the lock. This takes about 90 seconds.”

Trigger hard reset. After reset, try remote unlock again.

If still stuck after hard reset and remote unlock:

“I’m going to dispatch a technician to help with this. I know that’s not the answer you wanted — I’m really sorry. Can I get your name and a callback number? They’ll aim to reach you within [X minutes per regional SLA].”

Section A — Driver Actions That Are SAFE

Agents may suggest these while waiting for a technician:

1. **Lock and unlock the vehicle** — some vehicle systems re-query the inlet status on unlock, which can trigger a release

2. **Power cycle the vehicle** — turn off completely (not just accessory mode), wait 30 seconds, power back on. This resets the vehicle’s charging controller and may release the inlet latch.
3. **Check in-vehicle settings** — some vehicles have a “Unlock charge port” or “Stop charging” option in the infotainment system or in the manufacturer’s companion app
4. **Open and close the vehicle’s charge port cover** (if motorized) — cycling it may reset the inlet latch signal
5. **Vehicle emergency cable release** — see Section B

Section A — Actions That Are NOT SAFE

Agents must never advise these:

- **Pulling forcefully on the cable** — can damage the vehicle’s CCS inlet, connector latch, or the charger cable. Inlet repair costs \$1,000–\$3,000+.
 - **Using a tool (pry bar, screwdriver) on the connector** — will damage both connector and inlet.
 - **Disconnecting the charger from mains power** (at the panel) — not the driver’s job; creates safety hazards; may not release the lock on all charger models.
 - **Leaving the vehicle and driving away** — physically impossible with the cable locked, and attempting it will cause severe damage.
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Section B — Vehicle-Specific Manual Release Options

These are general guidelines. Always verify with the vehicle’s owner’s manual for the specific model year.

Tesla (NACS connector)

“On Tesla vehicles, there’s a manual cable release. Hold the rear driver-side door handle for about 3 seconds — that triggers an emergency release on some models. Alternatively, go to the charging menu on the screen and tap ‘Open Charge Port.’”

Tesla NACS connectors lock differently from CCS — they use a pin-actuated latch. If the vehicle is powered off and unresponsive, there may be a physical pull-tab behind the rear license plate or in the trunk (model-specific).

Hyundai / Kia (CCS)

Emergency manual release is located in the trunk or cargo area: > “In the trunk of your vehicle, there should be an emergency cable release — it looks like a small loop or handle near the charge port side of the trunk lining. Pulling it manually releases the inlet latch. Check your owner’s manual for the exact location.”

Chevrolet Bolt / Bolt EUV (CCS)

Manual release is in the front trunk (frunk) area or trunk: > “There’s an emergency charge port release cable in your vehicle — usually in the front storage area or trunk. It’s a small pull-cord that releases the port latch manually. Your owner’s manual will have the exact location.”

Ford Mustang Mach-E / F-150 Lightning (CCS)

Manual release accessible from inside the cargo area or via the Ford Pass app: > “Try using the Ford Pass app on your phone — there’s an option to unlock the charge port. If that doesn’t work, there’s a physical emergency release in the cargo area.”

Rivian (CCS / NACS)

Rivian has an emergency charge port release accessible via the Rivian app or from inside the vehicle: > “Open the Rivian app and look for the charge port unlock option. If the app isn’t working, there’s a manual release — check your vehicle’s settings screen under Charging, or consult the Rivian owner’s guide.”

Volkswagen / Audi / Porsche (CCS2 in Europe; CCS1 in US)

Some models have a manual release accessible from inside the vehicle: > “Check your vehicle’s infotainment system under Vehicle Settings > Charging — there may be an option to manually unlock the charge port. Your owner’s manual will also have a physical emergency release location.”

Nissan (CHAdMO on older models, CCS on newer)

“For CHAdMO vehicles like the LEAF, the charging port has a button on the charge port cover that can release the cable once the session is stopped. Try pressing the charge port release button on the vehicle while the session is confirmed stopped.”

Section C — Charger-Side Physical Release (Technician Action Only)

Most modern DC fast chargers have a manual override for the connector lock accessible to technicians — typically a small lever or button inside the charger cabinet behind a service panel. This is not accessible to drivers and requires a trained technician.

Do not instruct drivers to attempt to open the charger cabinet — it contains high-voltage equipment and is not safe for untrained personnel.

Section D — If the Driver Must Leave Urgently

If the driver has an emergency and cannot wait for a technician:

1. Confirm whether the vehicle has a tow mode or transport mode that can be activated while tethered (most do not, but check)
2. Contact local roadside assistance — some EV roadside services carry connector release tools
3. Contact the vehicle manufacturer’s roadside assistance line — they may have model-specific guidance
4. In absolute last resort, and only if the driver or a roadside professional does it: some connectors can be released by inserting a thin tool to manually depress the latch — this risks damage and should only be done by someone who knows the specific connector mechanism

“I strongly recommend waiting for our technician if you can — forcing the cable risks expensive damage to your car’s charging port. But if you need emergency help sooner, your vehicle manufacturer’s roadside line may be able to assist.”

Escalation Criteria

Condition	Action
Remote unlock fails after hard reset	Dispatch technician — priority
Driver has emergency and cannot wait	Contact regional tech dispatch for fastest ETA; also advise vehicle manufacturer roadside
Connector or inlet appears physically damaged	Dispatch + advise driver not to drive until inspected
Charger lost power during session (cable locked, charger off)	Dispatch — manual release required on charger side

Billing Note

If a driver was billed for time their vehicle was stuck at the charger (idle fees in particular), flag the session for billing review:

“I’ll make sure our billing team looks at this session — you won’t be charged idle fees for time the cable was stuck. Can I confirm your email for the review notification?”

Agent Reminders

- Stuck cable is an urgent situation — treat it with priority and empathy. The driver is stranded.
- Always try ending the session first before sending the unlock command.
- Never advise forcing the connector. The repair cost for a damaged CCS inlet can exceed the value of a month of charging.
- If dispatching a tech, give a realistic time estimate based on regional SLA — don’t say “shortly” without knowing.
- Log vehicle make/model for the technician — release procedures vary by vehicle.